Newton's Sums

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1 Introduction

Newton's sums, or Newton's identities, is a useful and efficient way to find the power sums of the roots of a polynomial without directly solving for the roots. Newton's sums is derived from Vieta's formulas, which relates the coefficients of the polynomial to its roots. For more information on Vieta's formulas, see this Brilliant article[1].

In this article, the following definitions will be used for simplicity. Do note that this notation is not universal, and if referenced in a formal proof, it should be followed with an explicit definition.

2 Notation

Let $f(x)$ be a polynomial, with $f \in \mathbb{Q}[x]$, of degree n and roots $r_1, r_2, \cdots, r_n$. Then,

Definition (Power Sum)**.** The k^{th} power sum P_k, where $k \in \mathbb{N}$, of the roots of $f(x)$ is the sum of each of the roots raised to the k^{th} power. In other words,

$$P_k = \sum_{i=0}^{n} a_i^k.$$

Definition (Elementary Symmetric Polynomial)**.** The j^{th} elementary symmetric polynomial S_j, with $j \in \mathbb{N}$ and $j \leq n$, of the roots of $f(x)$ is defined as

$$S_j = \sum_{1 \leq i_1 < i_2 < \cdots < i_j \leq n} r_{i_1} r_{i_2} \cdots r_{i_j}.$$

Experienced problem solvers will notice that these polynomials also show up in Vieta's formulas.

3 Motivation

Problem 1. *The polynomial $f(x) = x^2 - 3x + 2$ has roots r and s. What is P_n, for $n \in \mathbb{N}$?*

$n = 1$ is a straightforward application of Vieta's formulas. The sum of the roots is $r + s = 3$, so $P_1 = S_1 = 3$.

When $n = 2$, we can rewrite $r^2 + s^2$ in terms of S_1 and S_2. Note that

$$r^2 + s^2 = (r + s)^2 - 2rs,$$

[1] https://brilliant.org/wiki/vietas-formula/#:~~:text=x2%2Bbx%2Bc%E2%89%A1x2%E2%88%92(,q)x%2Bpq.&text=This%20is%20the%20so%Dcalled,to%20polynomials%20of%20higher%20degree.&text=x%202%20%E2%88%92%20(%20p%20%2B%20q,x%20%2B%20p%20q%20%3D%200.

and by Vieta's, this leads to $r^2+s^2=5$. Note that the roots of $f(x)$ are 1 and 2, and 1^2+2^2 indeed is 5.

Less obvious is how we express r^3+s^3 in terms of the symmetric sums. To get the cubic terms, we cube $r+s$, so

$$(r+s)^3 = r^3+3r^2s+3rs^2+s^3,$$

and the right side simplifies as $r^3+s^3+3rs(r+s)$. Thus $r^3+s^3=(r+s)^3-3rs(r+s)$, which equals 9.

As we continue raising S_1 to the nth power, there doesn't seem to be an easily recognizable pattern. For example,

$$\begin{aligned}
r^4+s^4 &= (r+s)^4-(4r^3s+6r^2s^2+4rs^3)\\
&= (r+s)^4-2rs(r^2+s^2+3rs)\\
&= (r+s)^4-2rs\left((r+s)^2+rs\right)\\
&= (r+s)^4-2(r+s)^2rs-2(rs)^2
\end{aligned}$$

and by a similar method,

$$r^5+s^5=(r+s)^5-5(r+s)^3rs-5(r+s)(rs)^2.$$

In particular, the coefficients follow no nice formula, and as n gets larger, the expansion of P_n becomes longer.

However, notice that in the expansion of P_4, we solved r^2+s^2 in terms of S_1 and S_2, which we have already done previously. Instead of expressing P_n solely in terms of S_1 and S_2, we can express it recursively, in terms of both S and P.

In $n=3$, consider other methods to get the cubic terms. Rather than cubing $r+s$, we can take $(r^2+s^2)(r+s)$. Then,

$$(r^2+s^2)(r+s)=r^3+s^3+rs(r+s),$$

so $r^3+s^3=(r^2+s^2)(r+s)-(r+s)rs$. By definition, this is also equal to $P_2S_1-P_1S_2$. Similarly, when $n=4$, we take

$$(r^3+s^3)(r+s)=r^4+s^4+rs(r^2+s^2),$$

or $P_4=P_3S_1-P_2S_2$. Continuing, we can solve for $P_5=P_4S_1-P_3S_2$ and $P_6=P_5S_1-P_4S_2$. In general, it seems that $P_k=P_{k-1}S_1-P_{k-2}S_2$ for non-negative integers k.

4 Generalization

We can extend this to higher degree polynomials. When we find the power sums of a cubic polynomial with roots r,s,t, a similar pattern occurs. We get

$$\begin{aligned}
r+s+t &= S_1\\
r^2+s^2+t^2 &= P_1S_1-2S_2\\
r^3+s^3+t^3 &= P_2S_1-P_1S_2+3S_3\\
r^4+s^4+t^4 &= P_3S_1-P_2S_2+P_1S_3
\end{aligned}$$

and so on. After applying a similar method to 4th, 5th, and even 6th degree polynomials, we arrive at a recursive generalization of the power sums known as Newton's Sums.

Theorem (Newton's Sums)**.** *For a polynomial $f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$ with roots $r_1, r_2, \cdots, r_n$,*

$$P_k = P_{k-1}S_1 - P_{k-2}S_2 + P_{k-3}S_3 - \ldots + (-1)^{n+1}P_{k-n}S_n$$

when $k > n$ and

$$P_k = P_{k-1}S_1 - P_{k-2}S_2 + P_{k-3}S_3 - \ldots + (-1)^{k+1}kS_k$$

when $k \leq n$, where $k \in \mathbb{N}$.

Proof. Consider the case when $k > n$. Take each root r_i, where i is an integer between 1 and n. By definition, $f(r_i) = 0$, and dividing by a_n in the polynomial expansion yields

$$r_i^n - S_1 r_i^{n-1} + S_2 r_i^{n-2} - \ldots + (-1)^n S_n = 0.$$

Notice that this is a direct application of Vieta's formulas. Now, we multiply each equation by r_i^{k-n}, giving

$$r_i^k - S_1 r_i^{k-1} + S_2 r_i^{k-2} - \ldots + (-1)^n S_n r_i^{k-n} = 0.$$

Summing each equation given by each of the roots and simplifying, we get

$$P_k = S_1 P_{k-1} - S_2 P_{k-2} + \ldots + (-1)^{n+1} S_n P_{k-n}.$$

Next, we consider when $k \leq n$. Note that

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_0 = a_n(x - r_1)(x - r_2)\ldots(x - r_n),$$

and dividing by a_n yields

$$x^n - S_1 x^{n-1} + S_2 x^{n-2} + \ldots + (-1)^n S_n = (x - r_1)(x - r_2)\ldots(x - r_n).$$

Taking the derivative of both sides with respect to x,

$$nx^{n-1} - (n-1)S_1 x^{n-2} + (n-2)S_2 x^{n-3} + \ldots + (-1)^{n+1} S_{n-1} = \sum_{i=1}^{n} \frac{\frac{f(x)}{a_n}}{x - r_i}.$$

Recall that $\frac{f(x)}{a_n} = x^n - S_1 x^{n-1} + S_2 x^{n-2} - \ldots + (-1)^n S_n$. To simplify the right hand side of the equation, we use the following lemma.

Lemma 1. *If r is a root of $f(x)$ and if $\frac{f(x)}{a_n} = (x - r)P(x)$, then the coefficient of the x^m term of $P(x)$ is*

$$(-1)^{n-m-1}S_{n-m-1} + (-1)^{n-m-2}S_{n-m-2}r + (-1)^{n-m-3}S_{n-m-3}r^2 \ldots + r^{n-m-1}.$$

for non-negative integers $m < n$.

The lemma can be proven using induction and synthetic division, and it is left as an exercise to the reader. Using lemma 1 and summing over all roots r_i, we get the polynomial

$$nx^{n-1} + (P_1 - nS_1)x^{n-2} + (P_2 - P_1 S_1 + nS_2)x^{n-3} + \ldots + (P_n - P_{n-1}S_1 + \ldots + (-1)^{n-1}nS_n).$$

Then, matching coefficients, we get

$$\begin{aligned}
-(n-1)S_1 &= P_1 - nS_1 \\
(n-2)S_2 &= P_2 - P_1S_1 + nS_2 \\
-(n-3)S_3 &= P_3 - P_2S_1 + P_1S_2 - nS_3 \\
&\ddots \\
(-1)^{n-1}S_{n-1} &= P_n - P_{n-1}S_1 + P + n - 2S_2 - \ldots + (-1)^{n-1}nS_n
\end{aligned}$$

After solving for the highest P_k, we get the desired formulas and we are done. □

5 Problems

Let's walk through a basic example.

Problem 2 (2003 HMMT February Algebra #7). *Let a, b, c be the three roots of $p(x) = x^3 + x^2 - 333x - 1001$. Find $a^3 + b^3 + c^3$.*

Solution. This is a direct application of Newton's sums. By the coefficients of $p(x)$, we know that $S_1 = -1$, $S_2 = -333$, and $S_3 = 1001$. Clearly, $P_1 = S_1 = -1$, and $P_2 = P_1S_1 - 2S_2$, which is equal to $(-1)^2 - (2)(-333) = 667$. Thus, we can use the following equation

$$P_3 = P_2S_1 - P_1S_2 + 3P_3,$$

which is $(667)(-1) - (-1)(-333) + (3)(1001) = \boxed{2003.}$ □

For more practice, here is a handful of applications of Newton's sums, not necessarily in increasing order of difficulty.

5.1 Practice Problems

Problem 3 (2008 HMMT February Algebra #1). *Positive real numbers x, y satisfy the equations $x^2 + y^2 = 1$ and $x^4 + y^4 = \frac{17}{18}$. Find xy.*

Problem 4 (2019 AMC 12A #17). *Let s_k denote the sum of the kth powers of the roots of the polynomial $x^3 - 5x^2 + 8x - 13$. In particular, $s_0 = 3$, $s_1 = 5$, and $s_2 = 9$. Let a, b, and c be real numbers such that $s_{k+1} = a\,s_k + b\,s_{k-1} + c\,s_{k-2}$ for $k = 2, 3, \ldots$. What is $a + b + c$?*

Problem 5 (2010 CHMMC Individual #5). *Let $x = \frac{3-\sqrt{5}}{2}$. Compute the exact value of $x^8 + \frac{1}{x^8}$.*

Problem 6 (2019 Purple Comet High School #28). *There are positive integers m and n such that $m^2 - n = 32$ and $\sqrt[5]{m + \sqrt{n}} + \sqrt[5]{m - \sqrt{n}}$ is a real root of the polynomial $x^5 - 10x^3 + 20x - 40$. Find $m + n$.*

Problem 7 (1985 IMO Longlist #88). *Determine the range of $w(w + x)(w + y)(w + z)$, where x, y, z, and w are real numbers such that*

$$x + y + z + w = x^7 + y^7 + z^7 + w^7 = 0.$$